Does Cross-Model Pollination Help Open Autoresearch? Evidence from Yukon Leaderboards

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Abstract

On open autoresearch leaderboards, many people point their own AI agents at one shared problem and build on each other's public submissions. A popular explanation for why this works is cross-model pollination: when someone using one company's model picks up work produced with a different company's model, the result is more likely to be an improvement. I tested this on the public record of 10 Yukon leaderboards, including ECDSA.Fail, covering 30,687 scored submissions. For each submission I recorded which company's model the participant declared and which earlier submission it built on. I fixed the whole analysis in advance, before the outcomes were joined to the data. The answer is no measurable effect. Raw promotion rates were 6.1% when building on another company's model and 7.6% when building on the same company's. Once each participant is compared with themselves at the same stage of a leaderboard, the chances are essentially equal. The odds are about 2% higher, and the data rule out anything beyond roughly 18% lower or 28% higher. That clears the pre-registered bar for calling the effect negligible on average, though a stricter bar set at first would have left it undecided. ECDSA.Fail alone has too few submissions to decide either way. The 843-qubit circuit quoted in the post that prompted this study did come from a cross-model step, which shows that such steps happen, not that they help on average. Code, data and the timestamped pre-registration are public.

1 Introduction

On 16 October 2026, Sreeram Kannan compared two circuit searches for elliptic-curve point addition in a public post [8]: "anons beat anthropic by 2x / anthropic gets 21M gates at 843 qubits / ecdsa.fail gets 11.9M gates at 843 qubits." The next day Kydo replied that this was "also a good case study of how cross modal pollination might produce better results" [10]. Figure 1 shows both posts.

The two searches differ in how they were organised. The first is the evolutionary framework of Gu et al. [4], in which a single model, Claude Mythos 5, proposes circuit-generating programs inside OpenEvolve and an independent evaluator scores them. One of its authors lists an Anthropic affiliation and another the Anthropic STEM Fellows Program. Its Table S3 reports a specialised Euclidean point adder at 843 qubits and, in Supplementary Section II.2, 20.5 million Toffoli gates. The second is ECDSA.Fail, an open leaderboard on which more than one hundred participants, each choosing their own models and tools, submitted improvements to a shared reversible circuit [12]. The two gate counts are not measured the same way. Table S3 of Gu et al. [4] counts reversible cleanup and excludes storage or lookup used to supply the classical addend. ECDSA.Fail scores the average executed Toffoli count over a submission-dependent set of Fiat–Shamir inputs [12, §6]. I use the comparison only as motivation, and nothing in this paper depends on it.

Kydo's reply names a mechanism that the ECDSA.Fail paper also describes, in prose, at four places. Its §4.1 says that "independent branches using different models, tools, backgrounds, and objectives also make the search resilient to plateaus." Its §4.3.2 reports that agents "from different

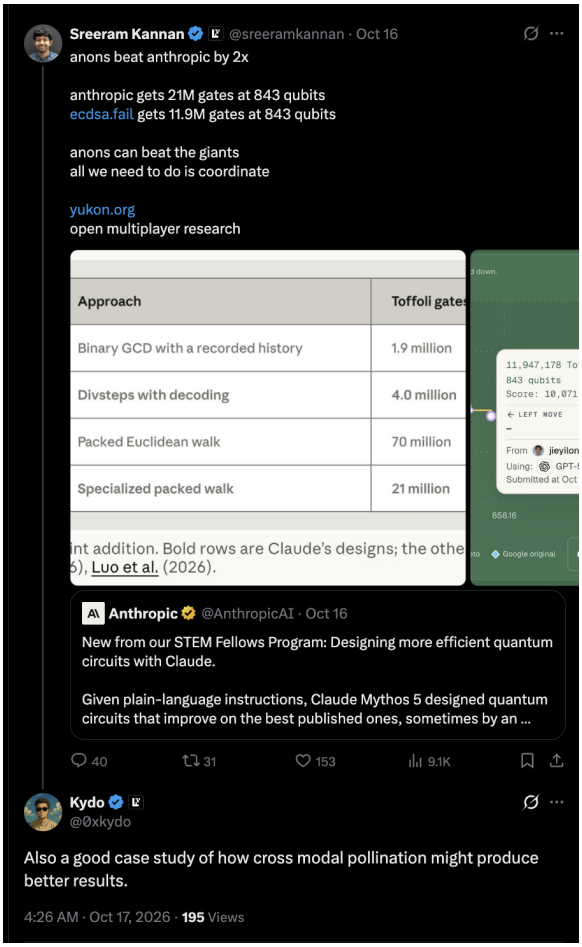


Figure 1: The post and reply that motivated this study, as displayed on X: the post by Sreeram Kannan on 16 October 2026, quoting Anthropic's announcement, and the reply by Kydo on 17 October 2026 [8, 10]. The 843-qubit, 11.9M-gate point it quotes is traced in Section 7.1.

model families (e.g. GPT, Claude, Grok, GLM, and Kimi) sometimes reviewed one another's conclusions," and that watcher agents, which compared adjacent public submissions and summarised the transferable techniques, "enabled rapid cross-pollination without collapsing the search onto a single trajectory." Its §4.3.3 states that "apparent plateaus were repeatedly broken by existing or newly joined contributors using different models, tools, expertise, or circuit families." Its §5.4.1 finds that "several plateau-ending advances came from participants other than those responsible for the preceding improvements." The same paper is explicit about what its record cannot show. In §6 it says the record does not "establish that one model, prompting strategy, or collaboration structure caused a given frontier movement." Its supplementary repository holds no model column in any data file; model names appear only inside the raw submission notes [11].

That leaves an open question that public data can address in part. When one participant builds on another participant's submission, does it matter whether the two used models from different vendors? I call this cross-model pollination. It differs from the people pollination of §5.4.1, where the new contributor is a different person but may use the same model. The ECDSA.Fail record does not separate the two, because an advance by a different participant can change the person, the model, or both.

Yukon, the platform that hosts ECDSA.Fail, makes the question testable [17]. Every submission carries a public note whose first line names the model the solver declared. Submissions also reference code commits in the benchmark's public repository, and some declare their parents through a structured attribution record. Yukon also hosts other benchmarks with the same submission format, several with thousands of submissions.

This paper is a follow-up test by one of the listed authors of the ECDSA.Fail paper. It takes the paper's prose observation as a hypothesis and asks whether the public record supports it once the person is held fixed. The analysis was pre-registered before outcomes were joined to treatments, and the result is reported whatever its sign. A null result has a defined meaning here, because the design includes an equivalence test with a margin fixed in advance.

The contributions are as follows.

- A frozen, checksummed snapshot of the public Yukon API, with a rebuild path that runs offline from it (Section 3).
- A dataset of every scored submission on 10 leaderboards chosen by a fixed rule, recording the company behind each declared model and the earlier submission each one built on (Sections 2 and 3).
- A test of whether building on another participant's work made with a different company's model changes the chance of success. It compares each participant with themselves, combines the leaderboards, and was timestamped before the outcomes were joined (Section 4).
- The answer: no measurable effect on average. Results follow for each leaderboard, nine alternative analyses, and case studies, including the history of the 843-qubit circuit quoted in the post above (Sections 5 to 7).

Section 2 describes Yukon and the study benchmarks. Section 3 describes the snapshot, labels, parents and outcomes. Section 4 gives the pre-registered design, a disclosure of what was seen before the lock, and a conflict-of-interest statement. Sections 5 to 7 report results, and Section 8 lists threats the design cannot remove. Section 9 discusses what the result means for running open autoresearch.

2 Setting

2.1 Yukon

Yukon describes itself as a platform for open frontier research, on which organisers create autoresearch challenges with verifiable benchmarks and solvers deploy agents against them [17]. Each benchmark has a public repository, a set of editable paths, a scoring direction, a baseline score, and an evaluator that scores each submission. The public API at `https://api.yukon.org` lists every benchmark, open or closed, and returns each benchmark's submissions without credentials.

A solver submits through the Yukon command-line tool. The tool requires a public Markdown note, a `--model` flag naming the underlying model, and a `--harness` flag naming the coding agent or harness [18]. It writes these two flags into the note as canonical `Model:` and `Harness:` lines. The tool's guidance asks for the exact, fully qualified model and tells solvers never to copy the model attribution from the base submission [18]. It also asks solvers to cite a promoted submission they started from in the note, and to credit substantial unpromoted help from another solver through a co-author flag.

The evaluator either accepts or rejects a submission, or the run fails. An accepted submission that improves on the current best is promoted onto the benchmark branch, and the landed commit is recorded as `promotedSourceRef` [18]. Each benchmark object carries a field `minScoreImprovementBips`, the smallest relative improvement that can promote, and the field differs between benchmarks. Normal synchronisation restores the editable paths from the best promoted submission, so the current best is the default starting point for new work [18]. A separate endpoint, `/api/submissions/{id}/attribution`, returns a structured attribution record for a submission. When the solver declared parents, the record has `declaration` set to `provided` and lists them as `dependencies`. Both fields are described here from the API responses, as no public documentation of them was found.

2.2 ECDSA.Fail

ECDSA.Fail, launched by Eigen Labs in late May 2026, optimises a reversible secp256k1 mixed point-addition circuit and scores it by $S = Q \times T$, the product of peak logical qubit width and average executed Toffoli count [12, §§3, 4.2]. Over about eight weeks, more than one hundred participants using diverse models produced over four hundred promoted submissions [12, §4.2]. Promoted improvements add their score, source, participant attribution and declared AI model to the public record [12, §4.2]. On Yukon the benchmark is `gpsanant/ecdsafail-challenge`. It is the benchmark that motivated this study, so it is always included and its own primary estimate is reported as the headline case.

2.3 Study benchmarks

The pre-registration fixes a selection rule, applied by `analysis/study.py` to the frozen snapshot and the frozen lineage files. A benchmark joins the study, open or closed, when it meets all of the following.

1. It has at least 500 analysis rows (Section 3.2), and at least 100 of them were promoted.
2. At least two model families each hold at least 10% of the analysis rows that carry a single known family.
3. Its trees are recoverable: a confirmatory window exists (Section 3.7), and at least half of its promoted analysis rows have a tree commit that the lineage build did not record as unfetchable.

ECDSA.Fail is included whatever the rule says.

Ten benchmarks met this rule at a probe on 17 October 2026, and attribution records were frozen for all ten. Table 1 lists them with their row counts in the frozen snapshot. The rule ran again on the frozen snapshot and lineage before the confirmatory analysis and returned the same 10 benchmarks. Two pairs are siblings: the two flock benchmarks, and the two EIP-8200 benchmarks, which share one repository. Sensitivity analysis 8 reduces each pair to one member (Section 4.5).

3 Data and measurement

This section explains how each submission becomes a data point: where the data came from, which submissions count, how a model is assigned to a company, how the parent is found, and how success is measured. The paper calls each submission a row.

Table 1: Study benchmarks. Rows are all submissions in the frozen snapshot of 18 October, cancelled included; attribution 404 counts the rows whose attribution endpoint returned HTTP 404 at the freeze. Source: `snapshot/FREEZE.md`.

Benchmark	Rows	Attribution 404
`davidtai/mlxfast-gemma4-26b-a4b`	3,239	397
`davidtai/qwen38-125b-a6b-cuda-v1`	5,757	337
`eigenlabs/eip8200-challenges/modexp`	1,512	151
`eigenlabs/eip8200-challenges/ripemd160`	1,233	122
`eigenlabs/flock-challenge`	4,632	53
`eigenlabs/flock-challenge-multi/x86`	3,392	166
`eigenlabs/lighter-prover-challenge`	8,379	338
`eigenlabs/mlxfast-challenge`	1,904	85
`eigenlabs/sig-golf/full`	933	246
`gpsanant/ecdsafail-challenge`	1,593	255
Total	32,574	2,150

3.1 Snapshot

The snapshot was fetched once from the public API without credentials, by `snapshot/fetch.py`. For every benchmark the API lists, the script stores the benchmark object and its full submission list as gzipped JSON with fixed timestamps. For the study benchmarks it also stores the attribution record of every submission, cancelled rows included. An attribution call that returns HTTP 404 is stored as unavailable, and the row's identifier is listed in the freeze record. After writing the files, the script writes `FREEZE.json`, with the fetch time and row counts, and a `SHA256SUMS` file covering every data file and `FREEZE.json` itself. Once a freeze exists, the script refuses partial refetches.

The fetch finished at 2026-10-18T13:35:31Z. It holds 48 benchmarks and 47,525 submissions, of which 32,574 belong to the ten benchmarks in Table 1 (`snapshot/FREEZE.md`). Of the attribution calls for those ten benchmarks, 2,150 returned HTTP 404. At the end of the freeze, a guard checked that every study benchmark holds exactly one attribution record per row, with no duplicates and none missing (`docs/JOURNAL.md`).

The loader, `snapshot/load.py`, reads only these files and checks them on every open (Appendix A.1). A submission is flagged as promoted when its status is `accepted` and its promotion status is `promoted`.

3.2 Which submissions count

Only submissions that the evaluator actually scored count, since a submission that never ran cannot succeed or fail on its merits. The paper calls these analysis rows, and `outcomes/denominator.py` decides which they are. The function drops cancelled rows and rows whose status is not accepted, rejected or failed. It drops accepted rows whose promotion status is set but is not one of promoted, failed or superseded, because their promotion was not final. It drops rows rejected before evaluation, matched by exact reason strings and patterns in `outcomes/denominator.yaml`. These are rejections on the solver's own claimed score, submissions with no code change, and packaging errors such as an archive path outside the editable paths or an over-long pull-request body.

Failed rows are classed by reason into three groups. Evaluated failures, where the scoring command ran or the submission's own code failed a gate or a build, are kept. Infrastructure

failures, such as runner setup, artifact transfer, dispatch errors, platform rate limits and interrupted validation jobs, are dropped. Ambiguous failures, whose text does not say whether the submission caused them, are kept, and sensitivity analysis 9 drops them. A failed row with no reason counts as ambiguous, and a reason that matches no class stops the pipeline with an error. The class list was built from the distinct failure reasons in the snapshot, without splitting any outcome by treatment.

3.3 Which company's model

Each submission's note starts with a line naming the model, and the study groups models by the company that makes them, called the model's family.

The declared model is the text after `Model:` on the first non-empty line of the note; a note whose first line is anything else has no declared model. A checked-in table, `families/table.yaml`, maps each exact declared string to a family, which is the model's vendor. The table holds 767 strings. Of these, 254 map to Unknown and 35 to Multi; the rest map to one of 16 vendor families (`families/table.yaml`).

The table was proposed by vendor keyword rules in `families/build_table.py` and then corrected by hand where note text gave evidence. A string naming two or more vendors is Multi, for example "Claude Opus 5.5 + GPT-6.1 Sol". A string naming two models from one vendor maps to that vendor. A model that a company names as its own maps to that company, as `Other:Cognition` for SWE-2 or `Other:Cursor` for Composer. A string maps to Unknown when it names an agent product with no model ("Devin", "Codex"), a name whose vendor is ambiguous ("Astra"), or something that is not a model at all. Strings that name the model being optimised rather than the model doing the work, such as target model weights on the inference benchmarks, are also Unknown. A row with no `Model:` line is Unknown. Lookup is exact: a declared string absent from the table raises an error, so every analysed string has a recorded decision.

Of the 32,574 study-benchmark rows, 32,386 carry a `Model:` line, and 1,453 of those, or 4.5%, map to Unknown. Among analysis rows, counting rows with no `Model:` line, 5.1% are Unknown. Unknown and Multi rows are excluded from the confirmatory tests. Sensitivity analysis 1 restores Unknown as a family of its own, and the exploratory analysis E1 studies Multi rows. Within a benchmark, a family with fewer than 30 analysis rows is folded into a single "Other" level for the family covariates of the model in Section 4.2. The `Harness:` line, when present, is kept for sensitivity analysis 6, because harness and family are close to collinear.

Label audit. A declared label is what the solver typed, so I measured how often the note body agrees with it. A seeded sample of 200 notes from benchmarks with at least 500 rows was drawn by `families/audit/sample.py` (seed 17). Five fresh-context Claude Opus 5.5 auditors each scored 40 notes as agree, contradict or silent against the `Model:` line, ignoring the header itself and verbatim copies of it. The result was 69 agree, 5 contradict and 126 silent (`families/audit/RESULT.md`). Agreement among non-silent notes was 69 of 74, or 93.2%, below the pre-registered bar of 95%. Four of the five contradictions change the family, three of them because a single-vendor label covered work that the body attributes to more than one vendor. At the family level, agreement was 70 of 74, or 94.6%. The auditors differed on one rule: three of the five batches counted an agent line naming the declared vendor's harness as agreement, and one counted it as silent. Recoding those 13 notes as silent gives 56 of 61, or 91.8%, at the model level. The audit is recorded, not a gate, so the shortfall is a limitation of the labels (Section 8).

3.4 What each submission built on

A submission's parent is the earlier submission it built on. Participants rarely say so in a machine-readable way, so the study uses three sources, from most to least direct. The lineage code in `lineage/` finds them from three sources, in a fixed order of precedence, and keeps the tree-distance parent for every analysis row as well.

Declared parents. When a row's attribution record has `declaration` equal to `provided` and lists dependencies, those are its declared parents. Declared identifiers are de-duplicated in order, and identifiers that are unknown or not created before the child are dropped and recorded. Declared parents are checked against every snapshot row of the benchmark, so a declared parent can be a row that is not itself an analysis row. One remaining declared parent gives a single-parent row. Two or more give a multi-source row, which is excluded from the confirmatory tests and studied in E1.

Bases named in the note. Many notes name their starting point on a `Base:` line. The code reads that line and resolves it to an earlier submission of the benchmark by identifier prefix, submission commit or promoted commit (`lineage/declared.py`). A resolved base is the parent of a row that has no declared attribution.

Tree-distance parents. Each analysis row has a tree: the commit in `promotedSourceRef` for promoted rows, and the commit in `submissionCommitSha` otherwise. The build fetches each commit by hash from the benchmark's public repository and records unfetchable ones in `missing.json` (Appendix A.2). The distance between two trees is the number of added plus removed text lines reported by `git diff --numstat --minimal --no-renames` over the benchmark's editable paths, with each changed binary file counted as one line. The candidates are the 50 most recent promoted rows whose promotion finished before the child was created (Appendix A.2). A solver's own unpromoted rows are not candidates. The tree parent is the candidate at the smallest distance, with ties going to the most recent row and then to the identifier.

Novel rows. A row is novel, and has no tree parent, when the text lines added relative to its nearest candidate exceed half of the text lines in its own tree. Binary files do not count toward size. Novel rows are excluded from H1 and H2 and reported descriptively.

Primary parent and freeze. The primary parent is the declared one when it exists, then the base named in the note when it resolves to an earlier row, and the tree parent otherwise. Sensitivity analysis 2 uses the tree parent for every row, which estimates a different quantity: the immediate code parent rather than the declared origin. The build writes, per benchmark, the tree manifests (path to blob hash), the distance table, the parent links and `missing.json` under `lineage/frozen/`, with their own `SHA256SUMS`. The analysis rebuilds from these files offline. The build refuses to overwrite a frozen benchmark when commits it already holds can no longer be fetched. The final build recorded no unfetchable commit on any study benchmark.

Validation. Acceptance test AT4 checks the inferred treatment cell, not the parent identifier, against two labelled sets: declared attribution edges, and notes that name a base in text. The bar is 85% agreement on text-declared bases; declared edges are reported but not gated, because only a few dozen of them can be checked. Agreement was 2,773 of 3,069 (90.4%) against text-declared

bases and 18 of 24 (75.0%) against declared edges. Split by whether the labelled parent was the current best when the child was created, agreement was 2,402 of 2,443 (98.3%) for current-best parents and 389 of 650 (59.8%) for the others. The parent rule took its final form after a first AT4 run failed before the lock. That run let a solver's own unpromoted rows compete as candidates even when a different base was named, and agreement on text-declared bases was 1,555 of 3,106, or 50%, mostly because a solver's own sibling attempt won the distance comparison (`docs/JOURNAL.md`). The revised rule, recorded in the specification before the lock, takes a declared parent first, then a base named in the note, and then the nearest promoted row.

3.5 The comparison groups

A row is cross-solver when its single primary parent was submitted by a different solver. It is cross-family when the parent's family differs from the child's, with both families known and single-vendor. Rows with a single parent and two known families fall into four cells: own lineage with the same family, own lineage with a switched family, another solver's work in the same family, and another solver's work in another family. The primary contrast fixes "another solver's work" and varies only the family. Within a solver, it compares that solver's cross-family attempts with the same solver's same-family attempts, which separates model pollination from people pollination.

3.6 Measuring success

The main measure of success is promotion: a submission is promoted when it beats the best score so far by the leaderboard's threshold and becomes the new default starting point.

The outcome code in `outcomes/` writes outcome columns only to a sealed directory, and the treatment dataset (`analysis/dataset.py`) carries no outcome column. Outcomes are joined to treatments only inside the confirmatory drivers, after the lock check passes (Section 4.6).

Rows are ordered by evaluation, using the time promotion finished and falling back to the last update time, because promotions can finish out of creation order. "Earlier" in every outcome below means earlier in this order. The running best before a row is the best promoted score among rows that finished strictly earlier, seeded with the benchmark's baseline score.

- Y1, promotion: the row's promotion status is `promoted`. Accepted rows whose promotion failed count as not promoted.
- Y2, Pareto gain (ECDSA.Fail only): the row has both a qubit count and a Toffoli count, and no earlier row with both counts dominates or equals its point.
- Y3, gain size: for a promoted row, the log ratio of the running best before it to its own score, signed so that larger is better under the benchmark's direction.
- Large jump: a promotion in the top decile of Y3 within its benchmark.
- Plateau break: a promotion that improves the running best by at least 1% and comes at least 72 hours after the previous promotion of that size, or after the benchmark's opening when there was none.

Recomputation. Acceptance test AT5 recomputes Y1 in evaluation order from the baseline-seeded running best, applying each benchmark's `minScoreImprovementBips`. Only accepted or rejected rows can promote or move the running best. Rows whose promotion status is failed or

superseded are named exceptions and are removed from the series. A benchmark whose recomputation disagrees with the recorded promotion status on more than 1% of its analysis rows is reported as "promotion rule not reproduced", and Y1 still comes from the recorded status. The recomputation reproduced the recorded promotions on 9 of 10 benchmarks; the rule was not reproduced on `eigenlabs/flock-challenge-multi/x86`. On `eigenlabs/flock-challenge-multi/x86`, every row that disagreed in a probe on 17 October predates 25 August 2026, the date of the first rejection citing a 100-basis-point threshold, so the threshold appears to have started part-way through the benchmark.

3.7 The traceable period

The chance of finding a submission's code must not depend on whether it succeeded, or the comparison would be skewed. On ECDSA.Fail, accepted rows lack a submission commit more often than rejected rows, mostly in June, while every promoted row has a promoted commit. Without a correction, promoted rows would have parents more often than other rows. For each benchmark, `lineage/window.py` groups analysis rows by calendar week, starting on Mondays in UTC. Walking back from the freeze, it finds the earliest week from which, through the freeze, the share of rows with a recoverable tree differs by at most 5 percentage points between promoted and other rows. The window starts at that week, and rows before it are reported descriptively only. Acceptance test AT15 checks the balance inside each window. On ECDSA.Fail the window starts in the week of 2026-06-08.

4 Pre-registered design

The design below was fixed in a written specification before any outcome was joined to a treatment, and the analysis code that implements it was locked with it (Section 4.6). Any change after the lock is recorded in `docs/DEVIATIONS.md` with its date and reason, and is reported in this paper. The file records no deviation: every confirmatory, secondary, sensitivity and exploratory result in this paper comes from the locked code. Descriptive numbers computed after the lock, such as the prediction interval in Section 5.2, come from separate scripts and are labelled where they appear.

4.1 Hypotheses

Every hypothesis is two-sided.

- H1 (primary): among cross-solver single-parent rows, cross-family rows have different odds of promotion (Y1) from same-family rows.
- H2: among own-lineage rows, a family switch changes the odds of promotion.
- H3a: among cross-solver promotions, large jumps are cross-family more often than other promotions.
- H3b: the same for plateau breaks.
- H4: among cross-solver promotions, gain size (Y3) differs between cross-family and same-family steps.
- H5 (ECDSA.Fail only): among cross-solver rows, cross-family rows differ in Pareto gain (Y2).

- E1 (exploratory): multi-family and multi-source rows differ from single-family, single-parent rows on Y1 and Y3.
- E2 (exploratory): the H1 effect varies over the life of a benchmark, by thirds of its analysis rows.

4.2 The main test

In plain terms, the main test asks whether a participant's submission is more likely to be promoted when it builds on another participant's work made with a different company's model than when it builds on another participant's work made with the same company's model. It compares each participant only with themselves, so that a skilled participant who happens to favour one kind of parent cannot create a difference on their own. It also adjusts for how far into the leaderboard's life the submission came, whether the parent was the best submission at the time, and how many submissions the participant had made before. The paragraphs below give the formal version.

The primary result is H1 pooled across all study benchmarks, ECDSA.Fail included. Each benchmark contributes one log odds ratio from a conditional logistic regression of Y1 on the cross-family indicator, stratified by solver [1]. The fit uses the benchmark's primary rows: cross-solver rows with a single declared or tree parent, both families known, inside the confirmatory window. Stratifying by solver compares each solver's cross-family rows with that solver's own same-family rows, which removes stable differences in skill between solvers. The covariates are:

- a restricted cubic spline in days since the benchmark opened, with five knots at the minimum, the three quartiles and the maximum of the benchmark's analysis rows, which gives four columns [5];
- child-family and parent-family indicators, after sparse families are folded;
- an indicator that the parent was the best promoted row whose evaluation had finished when the child was created;
- the logarithm of one plus the solver's earlier analysis rows on the benchmark.

The spline stands in for frontier difficulty and model releases over time. Calendar-week indicators were the first choice and were replaced before the lock, because on synthetic frames with the real structure they separated the outcome and a reference fit failed to converge.

Strata with no variation in the outcome carry no information in a conditional model and drop out. Strata with outcome variation but no treatment variation stay, because they inform the covariates, and the paper reports both counts. Before fitting, `analysis/models.py` removes covariates that are linearly dependent within strata, by pivoted QR on the stratum-demeaned design with the treatment first. If the treatment itself is dependent, the benchmark has no estimate. The conditional fit uses Newton's method and is checked for convergence through the score and for an unbounded likelihood. The locked code also rejects a fit in which any coefficient or standard error exceeds 15; the specification text does not state this rule, and Section 5.2 reports where it applied. If the data show separation or the fit fails these checks, Firth's penalised likelihood with solver indicators replaces it [3], and the paper says so for that benchmark. A fit that still does not converge is a failure, with no estimate, and is never reported as one. Planted-effect and placebo tests of the whole pipeline are described in Appendix A.4.

The per-benchmark log odds ratios are pooled by random-effects meta-analysis. The between-benchmark variance τ^2 is the DerSimonian–Laird estimate [2], with a Q-profile interval [16]. The

pooled interval uses the Hartung–Knapp–Sidik–Jonkman variance with a t distribution on $k-1$ degrees of freedom [6, 14]. ECDSA.Fail's own estimate is the headline case study and is also a secondary test.

4.3 How "no effect" is decided

Finding no difference only means something if the data could have shown one, so the plan fixed in advance a range of odds ratios that counts as "no meaningful effect", called the margin. If the result's 90% interval sits entirely inside that range, the effect is declared negligible; if the 95% interval excludes 1, an effect is declared; otherwise the result is undecided. Equivalence uses two one-sided tests at $\alpha = 0.05$ [13], which amounts to checking whether the 90% interval of the odds ratio lies inside the margin. The pre-registered margin is $[0.80, 1.25]$. The verdict is one of three.

- Effect detected: the 95% interval excludes an odds ratio of 1.
- Equivalent to no effect: the 90% interval lies inside the margin.
- Inconclusive: neither.

Both intervals come from the same distribution as the estimate. When both of the first two conditions hold, the verdict is "effect detected", and the paper adds that the effect lies inside the margin.

Power gate. Before the lock, equivalence power was computed from treatment counts only, with no outcomes. Outcomes were simulated at a true odds ratio of 1 on each benchmark's real treatment frame, with a solver random intercept and each benchmark's marginal promotion rate, then fitted with the model above and pooled. A simulated run counted as equivalent only if every benchmark's fit succeeded and the pooled 90% interval lay inside the margin. The pre-registered rule was to widen the margin to $[0.67, 1.50]$ if power fell below 80%. On ECDSA.Fail alone even an unclustered calculation gives $1.645 \times 0.137 \approx 0.226$, above $\log 1.25 \approx 0.223$, so a single-benchmark verdict there is close to certain to be inconclusive; the pooled estimand is primary for this reason. Over 2,000 simulated runs, power at the original margin was 0.6345, below the 80% bar, and 0.9905 at the widened margin, with a failed-fit share of 0.0005 across benchmark fits. The margin used in this paper is therefore $[0.67, 1.50]$, and `docs/prereg.md` states this reason under the values fixed at the lock. The widening rule was written into the specification on 17 October, before the snapshot was frozen.

4.4 Narrower questions

Besides the main test, the plan asks whether cross-company building changes the size of improvements, the chance of a large jump or of breaking a plateau, and, on ECDSA.Fail, the chance of a new Pareto point. Running many tests raises the chance of a false alarm, so those that qualify are corrected together.

Each secondary test is run per benchmark (`analysis/secondary.py`).

- H2 fits the primary model to own-lineage rows.
- H3a and H3b are permutation tests among cross-solver promotions. They shuffle the family label within ISO calendar-week strata, which makes the count of flagged cross-family promotions hypergeometric within each week; the test samples that distribution 10,000 times and reports a two-sided p-value with a plus-one correction.

- H4 is a median regression of Y3 on the cross-family indicator and the time spline [9], with a standard error from 2,000 bootstrap draws that resample solvers.
- H5 fits the primary model with Y2 as the outcome, on ECDSA.Fail only.

A secondary test is confirmatory only when the smaller of its two arms has at least 20 outcome events, counted after the lock and before the test result is computed. Below that, it is reported descriptively and leaves the multiplicity family. Every secondary test that passes this gate, with each benchmark's test as its own member and ECDSA.Fail's H1 included, forms one Holm family [7]. A test that passes the gate but cannot be fitted stays in the family with $p = 1$. E1 and E2 are exploratory and are reported without correction.

4.5 Alternative versions

Each of the nine alternatives below changes one judgment call and reruns the combined estimate.

Each sensitivity analysis recomputes the pooled primary estimate with one change (`analysis/sensitivity.py`).

1. Unknown is a family of its own instead of being excluded.
2. Every row takes its tree-distance parent, ignoring declared attribution.
3. Rows whose note says that only a nonce or a validation draw changed are dropped.
4. Families are coarsened to Anthropic, OpenAI and other.
5. The first 72 hours of each benchmark are dropped.
6. Harness indicators are added, each only when it adds rank beyond the family indicators.
7. The author's own rows (solver `owizdom`) are dropped.
8. Each sibling pair is reduced to one member, the one with fewer analysis rows.
9. Rows whose failure reason is classed ambiguous are dropped.

Analysis 3 targets nonce search, which the ECDSA.Fail paper identifies as a way to lower a reported score without changing the circuit [12, §6].

4.6 Lock and sealing

The lock binds the pre-registration text, the analysis code, the label and denominator tables, and the checksums of the frozen data to the git tag `prereg-v1`, and every analysis driver refuses to run unless the lock check passes (Appendix A.3). The commit of tag `prereg-v1` is `09c0815`, timestamped with OpenTimestamps [15] on 2026-10-19. The proof is committed as `docs/timestamps/prereg-v1.commit.ots`, and the confirmatory run happened after the tag.

4.7 What was seen before the lock

The analysis is not blind, and five things were computed before the lock.

1. On 16 October, I looked at the ECDSA.Fail leaderboard frontier and its low-qubit Pareto points.

2. On 16 October, I computed a crude version of H4 on ECDSA.Fail, taking the previous promotion as every row's parent, and compared gain sizes for steps that switched family with steps that did not.

3. On the same day, I computed a crude version of H3 on ECDSA.Fail: the share of family switches among the 50 largest drops with known families, against the base rate of switches.

4. On 17 October, I counted submissions, promotions and declared families on every benchmark the API lists. No outcome was split by family on any benchmark other than ECDSA.Fail.

5. On 17 October, a fresh-context reviewer computed a crude H1 and H2 promotion-rate split on ECDSA.Fail, taking the frontier holder at creation as the parent. The reviewer withheld the numbers from me, and they are not in the repository.

In item 2, 253 family-switching steps had a median gain of 0.063% and a mean of 0.306%, against 302 same-family steps with a median of 0.055% and a mean of 0.314%. In item 3, 16 of the 31 top-50 drops with known families switched family, against a base rate of 46%. My reading of both was that neither showed a visible difference on ECDSA.Fail, and the specification records the numbers verbatim. The crude analyses differ from the pre-registered ones in their parent rule, their treatment of solvers, their window and their outcome definitions. They touched only ECDSA.Fail, which is one of 10 benchmarks in the pooled primary.

4.8 Conflict of interest

I am a participant in the system I study. I am solver `owizdom` on ECDSA.Fail, with five submissions on 2 June 2026, three of which were promoted, and I have no rows on the other study benchmarks. I am also a listed author of the ECDSA.Fail paper whose prose claims this study tests [12]. Sensitivity analysis 7 drops my rows from every benchmark. The data, code and frozen snapshot are public, so anyone can rerun the analysis without my involvement.

Use of AI tools. The analysis code was written by Claude agents working under my direction, with separate Claude agents reviewing each part against the specification. The label audit in Section 3.3 was scored by Claude Opus 5.5. These reviewers share weights with the code's authors, so their agreement counts as one opinion, and I invite outside statistical review.

5 Results

The short version: building on another company's model neither helps nor hurts, on average. The rest of this section gives the numbers behind that sentence. Every number is generated from `results/*.json`, either by the locked analysis code (`analysis/numbers.py`) or, for descriptive extras such as raw rates and the prediction interval, by `tools/paper_numbers.py`, which sits outside the lock.

5.1 Who builds on whom

The 10 leaderboards hold 30,687 scored submissions. The main analysis keeps only submissions that built on one earlier submission by a different participant, with a known company on both sides, inside the period where parents can be traced fairly (Section 3.7). Five reasons remove the rest, counted once each in a fixed order:

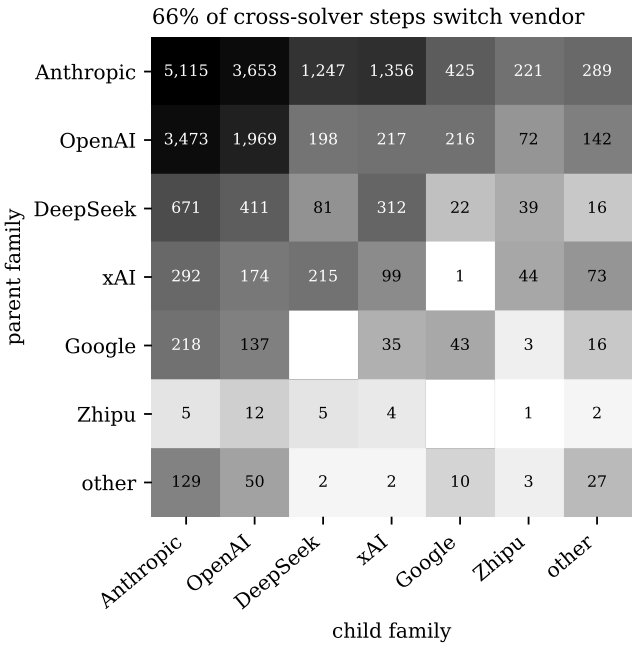


Figure 2: Who builds on whom. Each cell counts submissions that built on another participant's work, by the company behind the parent's model (rows) and the new submission's model (columns), over the 10 leaderboards. Off-diagonal cells are cross-company steps. The six largest companies are shown; "other" groups the rest, so its diagonal cell mixes companies. Source: `analysis/dataset.py`; figure by `tools/paper_figures.py`.

- 5,085 built on the participant's own earlier work (these are used for H2);
- 190 declared more than one parent;
- 1,135 were new enough code to have no parent, or had no candidate parent;
- 2,530 had an unknown or mixed company on one side;
- 110 came before the traceable period.

That leaves 21,637 submissions. Participants whose submissions all succeeded or all failed tell the model nothing about the difference, so the fits use the remaining 18,383.

Most parents were found from the code itself: 84.7% by comparing code trees, 10.7% from a base named in the note, and 0.2% from Yukon's structured attribution record.

Building on another company's model is the normal case on these leaderboards. Among all submissions with one parent and known companies, 14,433 built on another participant's work made with a different company's model, and 7,314 on another participant's work made with the same company's. Within a participant's own line of work, switching company was rare: 403 switches against 3,631 that kept the same company. Figure 2 shows the flows between companies; Anthropic and OpenAI dominate both sides, and the largest cross-company flows run between them.

5.2 The main result

Raw promotion rates point slightly against cross-company building. Of the 14,379 submissions that built on another company's model, 880 were promoted (6.1%). Of the 7,258 that built on the same company's model, 552 were promoted (7.6%). These raw rates mix together who submitted, when, and from which starting point, so they are not the test.

The pre-registered test compares each participant with themselves: their cross-company attempts against their own same-company attempts, at the same stage of the leaderboard, with the same starting-point information (Section 4.2). Done separately on each leaderboard and then combined, it gives an odds ratio of 1.023. In words, the odds of promotion are about 2% higher when building on another company's model, which is no real difference. The 90% interval runs from 0.821 to 1.276: the data rule out a disadvantage bigger than about 18% or an advantage bigger than about 28%. The 95% interval is 0.780 to 1.343.

The plan fixed in advance that an effect counts as negligible if the 90% interval sits entirely inside 0.67 to 1.50. It does, so the pre-registered verdict is *equivalent to no effect*. Figure 3 shows each leaderboard and the combined estimate.

That bar was not the first one. The plan began with a stricter bar of 0.80 to 1.25, and under it the result would be undecided, because the upper end, 1.276, sits just above 1.25. Before any outcome was joined to the data, a simulation using only the structure of the data showed that the stricter bar would reach a verdict only 0.6345 of the time even if there were truly no effect, against 0.9905 for the wider bar. A rule written on 17 October, before the data were frozen, said to switch to the wider bar in that case, and the switch is recorded in the locked plan (Section 4.3).

Two further points sharpen what the verdict covers. First, it is about the average across leaderboards. Leaderboards differ a little from each other (between-leaderboard variance $\tau^2 = 0.063$), and a 90% prediction interval for the effect on a new leaderboard runs from 0.61 to 1.72. So the data are consistent with no effect on average while leaving room for modest effects, either way, on a single leaderboard; this interval was not part of the plan. Second, every leaderboard was fitted with Firth's method, a standard correction for small or lopsided groups, instead of the planned conditional model. On eight leaderboards the reason was the one the plan names: some rarely used companies had no promoted submissions at all, or only promoted ones, which breaks the planned model. On `davidtai/qwen38-125b-a6b-cuda-v1` and `eigenlabs/eip8200-challenges/modexp` the switch came from a safety rule in the locked code, not stated in the plan's text, that rejects any fit with a coefficient or standard error above 15; the time-trend terms crossed it there. The planned model never failed on the cross-company comparison itself. The estimates follow the locked code, and `docs/DEVIATIONS.md` records the mismatch.

5.3 ECDSA.Fail on its own

On ECDSA.Fail alone the odds ratio is 0.408, from 417 submissions and 20 informative participants. The 95% interval runs from 0.138 to 1.205 and the 90% interval from 0.16 to 1.01. The best guess points against cross-company building, but the interval is wide enough to include anything from a large disadvantage to a small advantage, so the verdict is *inconclusive*. This was expected: the plan noted before the lock that ECDSA.Fail alone is too small to settle the question (Section 4.3), which is why the main test combines all ten leaderboards. As one of the secondary tests, its p-value is 0.105, or 1.000 after correcting for the number of tests.

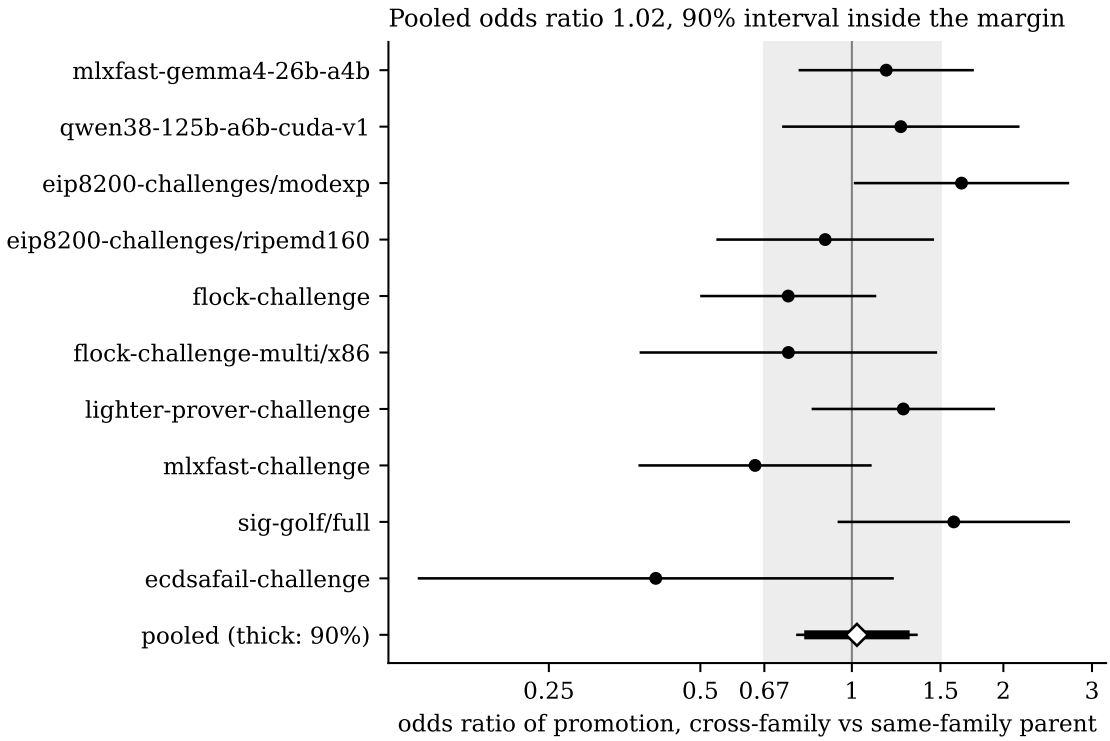


Figure 3: The main result. Each dot is one leaderboard's odds ratio of promotion for building on another company's model against the same company's, with its 95% interval; 1 means no difference. The diamond is the combined estimate with its 95% (thin) and 90% (thick) intervals. The shaded band, 0.67 to 1.50, is the pre-registered range for "no meaningful effect". Source: `results/confirmatory.json`; figure by `tools/paper_figures.py`.

5.4 Secondary questions

The plan also asked four narrower questions (Section 4.4); Table 4 in Appendix B lists every test. A test only counts when the smaller group has at least 20 successes, so that it is not decided by a handful of events. 12 tests passed that bar: ECDSA.Fail's main test, the size-of-gain test (H4) on each of the 10 leaderboards, and the Pareto test (H5) on ECDSA.Fail. After correcting for running that many tests, none shows an effect; the smallest corrected p-value is 0.154. Switching company within your own line of work (H2) was too rare, and the large-jump and plateau-break questions (H3a, H3b) had too few events, to be tested on any leaderboard, so they are reported only as description. Large jumps by other participants number between 1 and 10 per leaderboard in the smaller group, and plateau breaks at most one.

One leaderboard, `eigenlabs/eip8200-challenges/modexp`, has a 95% interval that excludes 1 (uncorrected $p = 0.043$). The per-leaderboard estimates are pieces of the combined estimate, not separate tests, and with ten leaderboards about one interval in twenty, so half of one, is expected to miss 1 by chance.

Table 2: Per-benchmark primary results. Names omit the setter prefix of Table 1. Analysis rows are all scored submissions; fit rows are primary rows from solvers with outcome variation. Strata kept are informative solvers; the next column counts those with no treatment variation. Odds ratios carry 95% intervals. Source: `results/confirmatory.json`.

Benchmark	Analysis	Fit	Strata	No treat.	Method	Odds ratio [95%]
`mlxfast-gemma4-26b-a4b`	2,813	2,129	40	4	Firth	1.17 [0.79, 1.74]
`qwen38-125b-a6b-cuda-v1`	5,409	4,095	21	2	Firth	1.25 [0.73, 2.14]
`eip8200-challenges/modexp`	1,332	846	20	2	Firth	1.65 [1.02, 2.69]
`eip8200-challenges/ripemd160`	1,073	545	19	0	Firth	0.89 [0.54, 1.45]
`flock-challenge`	4,605	1,706	36	9	Firth	0.75 [0.50, 1.11]
`flock-challenge-multi/x86`	3,240	2,154	19	6	Firth	0.75 [0.38, 1.47]
`lighter-prover-challenge`	8,208	5,322	46	4	Firth	1.27 [0.84, 1.91]
`mlxfast-challenge`	1,883	890	32	8	Firth	0.64 [0.38, 1.09]
`sig-golf/full`	685	424	23	2	Firth	1.59 [0.94, 2.70]
`ecdsafail-challenge`	1,439	272	20	4	Firth	0.41 [0.14, 1.20]
Pooled (HKSJ)	30,687	18,383				1.023 [0.780, 1.343]

5.5 Exploratory analyses

These were not part of the main plan and carry no correction. E1 compares the 339 submissions that mixed companies or declared several parents with the 27,618 that did not; most leaderboards have too few mixed submissions for a stable estimate, and the results are in `results/exploratory.json`. E2 asks whether the effect changes over a leaderboard's life by splitting each one into thirds by time. The combined odds ratios are 1.126, 0.900 and 1.041 for the first, middle and last third, all close to 1.

6 Each leaderboard, and alternative analyses

6.1 Each leaderboard

Table 2 gives each leaderboard's numbers. The leaderboards do not lean one way: five estimates lie above 1 and five below, ranging from 0.408 on `gpsanant/ecdsafail-challenge` to 1.651 on `eigenlabs/eip8200-challenges/modexp`. Between 19 and 46 participants per leaderboard had both successes and failures, which is what the comparison needs. On nine leaderboards some of those participants only ever built on one kind of parent, so they help with the time trend and other adjustments but not with the comparison itself.

As a data check, I recomputed which submissions should have been promoted from their scores, and the recomputation matched Yukon's records on 9 of 10 leaderboards (Section 3.6). On `eigenlabs/flock-challenge-multi/x86` it disagrees on more than 1% of submissions, apparently because the promotion threshold changed part-way through, so the analysis uses Yukon's recorded promotions there, as the plan requires.

6.2 Alternative analyses

A result can depend on judgment calls, so the plan fixed nine alternative versions of the analysis in advance, each changing one call (Section 4.5). Table 3 and Figure 4 show them. Every version gives a combined odds ratio between 1.01 and 1.03, and every 90% interval stays inside the pre-registered range, so no single judgment call changes the answer. Version 2 asks a slightly different question,

Table 3: Pooled odds ratio under each sensitivity analysis, with the number of benchmarks pooled and 95% and 90% intervals. Source: `results/sensitivity.json`.

Analysis	k	Odds ratio [95%]	90% interval
Primary	10	1.02 [0.78, 1.34]	[0.82, 1.28]
1. Unknown as a family	10	1.03 [0.79, 1.34]	[0.83, 1.28]
2. Tree parent for every row	10	1.02 [0.78, 1.33]	[0.82, 1.26]
3. Drop nonce-only rows	10	1.01 [0.76, 1.35]	[0.80, 1.28]
4. Coarse families	10	1.01 [0.77, 1.33]	[0.81, 1.26]
5. Drop first 72 hours	10	1.01 [0.75, 1.37]	[0.79, 1.29]
6. Harness indicators	10	1.02 [0.78, 1.32]	[0.82, 1.26]
7. Drop the author's rows	10	1.02 [0.78, 1.34]	[0.82, 1.28]
8. One per sibling pair	8	1.02 [0.75, 1.38]	[0.79, 1.30]
9. Drop ambiguous failures	10	1.03 [0.78, 1.36]	[0.83, 1.29]

whether the immediate code parent matters rather than the declared starting point, and agrees too. Version 7 is identical to the main result: the author's 5 submissions are all on ECDSA.Fail, and the three that are analysis rows predate its confirmatory window. Version 8 keeps one leaderboard from each pair of closely related leaderboards and combines eight.

Unknown model labels are concentrated on one leaderboard. On `eigenlabs/flock-challenge`, 1,416 of 4,605 analysis rows are cross-solver rows with an Unknown or Multi family on one side, and in 1,140 of them it is the parent's family that is missing. Version 1, which treats Unknown as a company of its own, moves that leaderboard's odds ratio from 0.747 [0.503, 1.111] to 0.79 [0.53, 1.16]. This per-leaderboard figure is descriptive; the plan reports only the combined value for version 1.

7 Case studies

The cases below are read from public notes and API fields in the frozen snapshot, each cited by submission identifier. They show how the measurement behaves on individual rows, and none of them is evidence for or against H1.

7.1 The 843-qubit point

The post that motivated this study quotes 11.9 million gates at 843 qubits for ECDSA.Fail [8]. The snapshot holds that point as submission `4a8ade44` by solver `jieyilong`, created on 11 October 2026, with 843 qubits and 11,947,178 average executed Toffoli gates. Its declared model is GPT-5, and its note names the coding agent as Codex with separate builder and auditor roles. Figure 5 shows the chain behind it as the notes describe it.

The note says the submission "starts from gnuchev's public Froghop-One submission `3f1c7ce`", at 843 qubits and 13,010,600 Toffoli gates. That row, `3f1c7ce5`, was created on 9 October and declares Claude Opus 5.5 in Claude Code. Its own note names its base as gnuchev's earlier 849-qubit circuit, built on two rows by BitWonka from the same day, both declared Claude Opus 5.5. The jieyilong note makes two changes at the same qubit count. It says the second transfers a measurement and phase-repair pattern from BitWonka's Frogstrip line and from jieyilong's own 975-qubit submission `ba4fda87`, declared GPT-5 Codex. It also says that its builder and auditor roles "used the same model family; no model-diversity claim is made."

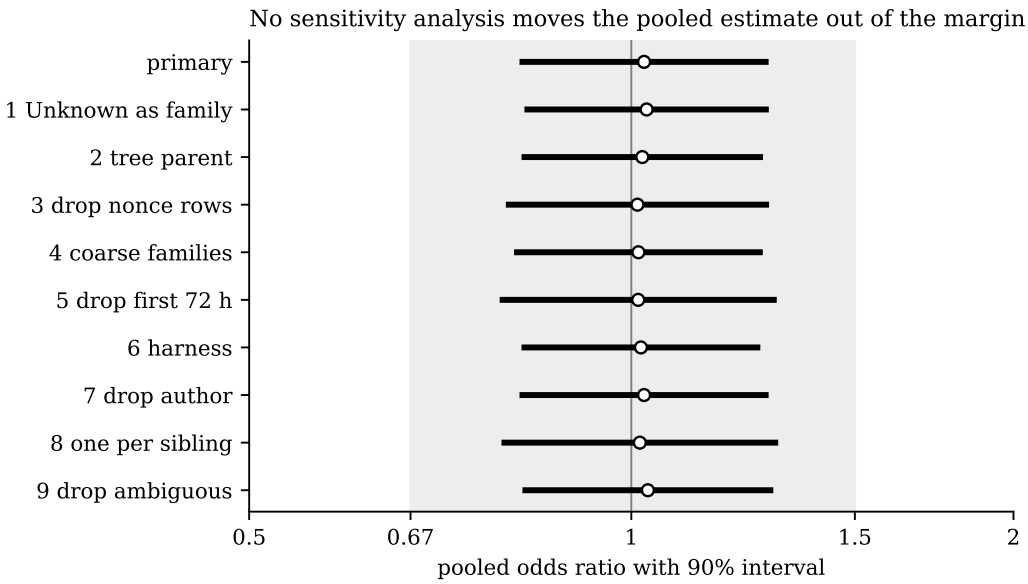


Figure 4: Pooled odds ratio and 90% interval under the primary analysis and each sensitivity analysis. The shaded band is the margin. Source: `results/sensitivity.json`.

By the notes, then, the quoted point is a cross-solver, cross-family step: an OpenAI model building on another solver's Anthropic-model circuit. Within each solver the chain is single-vendor, and the step that produced the point names one family. This is the pattern Kydo's reply describes, and it is one step.

The case also shows two limits of the measurement. First, none of these rows promoted. Promotion on ECDSA.Fail follows the product of qubits and Toffoli gates, and the best promoted product just before `4a8ade44` was 943,306,548, about 10.7 times lower than its own. The 843-qubit rows are low-width Pareto points, so under Y1 the step the post quotes is a failure, and only H5 on Pareto gain could count it. Second, the tree-distance rule did not find the base the note names. Its candidates are promoted rows only (Appendix A.2), so an unpromoted row such as `3f1c7ce5` can never be a tree parent. The note also names its base in prose rather than on a `Base:` line, which is the only form the text rule reads. The rule assigned `dfb1aec2`, a promoted Claude Opus 5.5 row by leech1996. The parent identifier is wrong, but the treatment cell is the same, cross-solver and cross-family, because both candidate parents declare an Anthropic model. AT4 measures exactly this agreement of cells rather than identifiers. On 17 October, after the post and before the freeze, solver `jackylee0424` reached 843 qubits with 9,382,317 Toffoli gates, declaring Claude Opus 5.5; that row was not promoted either.

7.2 The largest gains

The largest promotion on ECDSA.Fail by Y3 is `0c1d4d95` by `pldallairedemers` on 2 June, declared GPT-5, which lowered the best score by 18.8%. The tree rule classes it as novel, with no parent, and it predates the confirmatory window, so it is outside the primary sample.

The two largest promotions inside the window point in opposite directions. On 21 August, `3616dbf2` by `teddyjfpender`, declared GPT-5.6 Sol in a Codex multi-agent loop, lowered the best score by 14.8%. It replaced the point-addition circuit with one built around a fixed-depth ping-pong

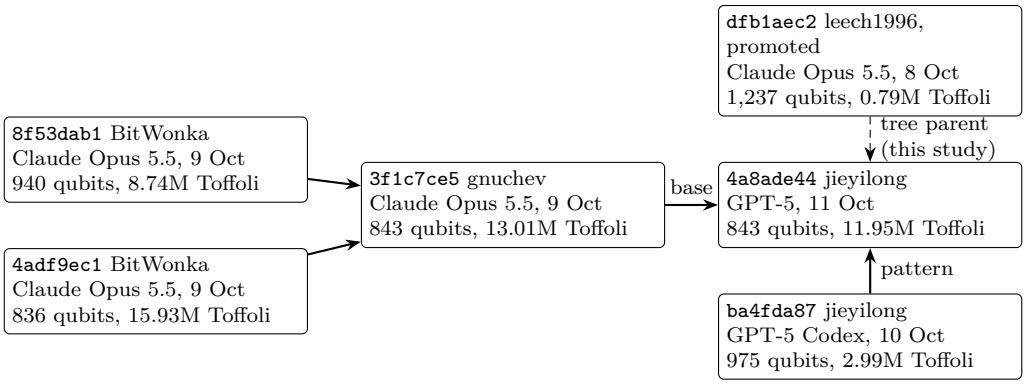


Figure 5: The chain behind the 843-qubit, 11.9M-Toffoli point on ECDSA.Fail. Solid arrows are bases and transferred patterns named in the notes, through gnuchev's own 849-qubit step, which is not drawn. The dashed arrow is the parent that the tree-distance rule assigns. None of the rows except `dfb1aec2` was promoted. Source: frozen snapshot notes and `lineage/frozen`.

binary division. Its tree parent is a promoted row by `jackylee0424` declared GPT-5, so it counts as a same-family step. On 7 October, `53986ea4` by BitWonka, declared Claude Opus 5.5 with Claude sub-agents, lowered the best score by 9.7% with a new division walk. Its note names its base as the "GCD Skywalk line", which is the promoted row `73c5e2a8` by `jackylee0424`, and the tree rule assigns the same row. That parent declares GPT-6.1, so the step counts as cross-family.

The parent's own note complicates the label. It credits GPT-6.1 Sol in Codex for the final search and packaging, and Claude Opus 5.5 in Claude Code, together with earlier Codex agents, for route and formal-lever development. Under its single-vendor label, the parent is OpenAI work; by its own description it is work by two vendors. This is the most common disagreement in the label audit (Section 3.3): a single-vendor label on multi-vendor work blurs the line between the two arms in a way the `Model:` line cannot show. Large gains occur on both sides of the treatment: the two cases here share a parent family, OpenAI, and differ in the child's family.

8 Threats to validity

The design removes some sources of bias and leaves others in place. This section lists the ones it cannot remove, in roughly the order of how much I expect them to matter.

Self-selection within solvers. Participants choose both their parents and their models. Solver strata remove stable traits, such as skill or compute budget, from the primary contrast. They do not remove choices that vary within a solver. A solver who reaches for a different model when a problem looks harder, or who builds on another solver's work only after a promising idea appears, changes the comparison in ways the covariates may not capture. The time spline and the current-best indicator absorb part of this, and the rest remains.

Self-declared labels. The `Model:` line is what the solver typed, and the platform does not verify it. In the audit, 93.2% of non-silent notes agreed with the declared model and 94.6% with the declared family, below the 95% bar set in advance (Section 3.3). Most disagreements were multi-vendor work under a single-vendor label, which puts rows in one arm when the work belongs partly to both. Uniform label noise of this kind tends to pull an odds ratio toward 1, so it bears on

how to read an equivalence verdict. Sensitivity analyses 1, 4 and 6 change how labels and harnesses enter the model. The audit itself was scored by models from one of the vendors being labelled.

Parent inference. Declared attribution exists for only some rows, and the tree-distance parent is an inference from code similarity. A solver can build on an idea without copying its code, and two solvers can reach similar code independently. AT4 measures how often the inferred cell matches the cell implied by declared or text-named parents (Section 3.4). Agreement is high when the labelled parent was the current best and much lower otherwise, so rows that build on older work carry more treatment error. Sensitivity analysis 2 replaces every declared parent with the tree parent. Rows whose tree is far from every candidate are treated as novel and leave the primary sample, which changes who is compared with whom.

Missing attempts. The API holds only submitted work. Local experiments, discarded branches and private agent sessions are absent, a point the ECDSA.Fail paper also makes about its own archive [12, §6]. Y1 is therefore promotion per evaluated submission, never per attempt. If solvers submit cross-family work only when it looks promising, the per-submission rate overstates its per-attempt rate, and the same holds for same-family work.

Missing trees. On ECDSA.Fail, commit availability differs between accepted and rejected rows in the early weeks. The confirmatory window starts only once tree recovery is balanced between promoted and other rows, and AT15 checks this. Rows before the window are excluded, so results describe the later part of each benchmark.

The denominator. The line between an evaluated failure and an infrastructure failure was drawn from reason text. Ambiguous reasons are kept in the primary analysis and dropped in sensitivity analysis 9.

Dependence between rows. Resubmissions of one idea, shared parents and repeated nonce draws make rows within a benchmark dependent in ways the solver strata do not fully capture. Sensitivity analysis 3 drops rows whose note reports only a nonce or validation-draw change. Sibling benchmarks share tasks or repositories, and sensitivity analysis 8 keeps one member of each pair.

One platform. Every benchmark runs on Yukon, with one submission format and one promotion mechanism. Promotion thresholds differ between benchmarks, and on at least one benchmark the threshold appears to have changed part-way through (Section 3.6). Results may not carry over to settings where participants coordinate differently or share work outside a leaderboard.

The motivating comparison. The gate counts in the 16 October post come from two searches with different interfaces and accounting conventions (Section 1). The study does not use that comparison as evidence, and its result says nothing about which search found the better circuit.

What was seen before the lock. Crude analyses on ECDSA.Fail were computed before the lock (Section 4.7). The parent rule, sample and model were fixed in writing before outcomes were joined, and the lock binds the code to that text. A reader can still discount ECDSA.Fail's own estimate on these grounds, and the pooled primary is less exposed because the other benchmarks were not split by family before the lock.

Conflict of interest. I am a participant on ECDSA.Fail and a listed author of the paper whose claims this study tests (Section 4.8). Sensitivity analysis 7 removes my rows, and the full pipeline is public.

Reviewer independence. The code and the label audit were reviewed by Claude models, which share weights with the agents that wrote the code. Agreement among them is one opinion, and the design has not yet had an outside statistical review.

9 Discussion

9.1 What the result means

The plain reading is that the brand of model behind the work you build on does not matter for whether your submission gets promoted. Across ten Yukon leaderboards, a participant who builds on another company's model succeeds about as often as when the same participant builds on the same company's model. The data rule out a difference bigger than roughly 18% down or 28% up in the odds. Every alternative version of the analysis lands within 0.02 of the main estimate. For someone choosing which public submission to start from, picking one made with a different company's model gains nothing on average, and loses nothing either.

Three limits apply. The result is an average: on a single new leaderboard the effect could plausibly sit anywhere from 0.61 to 1.72. It counts only submitted work, so experiments that never reached the leaderboard are invisible. And it compares each participant with themselves, both times building on someone else's work, so it isolates the model and holds the person fixed. A sharper answer, one that would pass the original stricter bar, needs more data than these ten leaderboards hold.

The labels matter too. If participants' declared models were often wrong in random ways, the estimate would be pulled toward "no difference", making this kind of verdict easier to reach than it should be. The label audit found that 93.2% of the notes that mention a model agree with the declared one (Section 3.3), so the pull is limited, but it is not zero.

9.2 Relation to the ECDSA.Fail paper

The ECDSA.Fail paper describes cross-pollination in prose at four places [12], quoted in Section 1. Its §5.4.1 observes that the advances that ended plateaus often came from participants other than those behind the previous improvements. This study holds the participant fixed and asks only about the model, so it tests a narrower claim than that paper makes. The two fit together: different people can matter even if different models, as declared, do not. Its §4.1 and §4.3.3 attribute resilience to branches that differ in "models, tools, backgrounds, and objectives"; a submission record can separate only the declared model from that list. Its §4.3.2 describes agents from different families reviewing one another's conclusions inside one participant's workflow. That kind of pollination is invisible in submission data, because it leaves one `Model:` line, and the case in Section 7.2 shows a note that credits two vendors under a single-vendor label. Plateau breaks, the events those passages emphasise, were too rare to test on any leaderboard.

Kydo's reply used the phrase "cross modal", which in machine learning usually means mixing modalities such as text and images. Read in context, it refers to models from different vendors, and this paper uses that reading throughout.

9.3 What else could explain the 2x

The comparison in the motivating post [8] is between one model in an evolutionary loop [4] and an open swarm of participants [12]. This study finds that, inside the swarm, which companies' models meet in a parent and child does not predict success. That points to the swarm's other features. Many independent people bring their own tools and goals, everyone restarts from a public best-so-far, and an evaluator checks every submission around the clock. That reading fits the post's own conclusion, that what open research needs is coordination. Every new attempt on Yukon starts by default from the best promoted submission (Section 2), and the primary model holds that choice fixed through its current-best covariate. Which of these features drove the difference between the two searches is a separate question, and the data here do not answer it.

9.4 What a platform would need to log

To answer whether mixing models causes better results, rather than whether it goes with them, a platform would need data that a submission record does not hold.

- Every attempt, not only submissions, so that selection into submission can be measured.
- The model and harness of each agent that touched an attempt, not one declared line per submission.
- Declared parents on every submission; here declared attribution set the parent of only 0.2% of analysis rows.
- Agent session provenance that ties code changes to the model that wrote them.
- For a direct test, randomised model assignment on some attempts, which an organiser could offer as an opt-in track.

Yukon already records traces privately and links them to submissions [18], so the first and fourth items may exist for the organisers even though they are not public.

9.5 Limits

Section 8 lists the threats in full. The most important are that participants choose their own models and starting points, and that model labels are self-declared. Another is that parents found from code put a submission in the same group, cross-company or same-company, as the parent the participant named 98.3% of the time when that parent was the current best, but only 59.8% of the time otherwise. The code and the label audit were reviewed by Claude models, and the analysis would benefit from outside statistical review. The release that accompanies this paper holds the snapshot, the lineage files, the code and the pre-registration, so every number here can be rebuilt offline.

Data and code availability

The paper is archived at `https://doi.org/10.5281/zenodo.23465602`. The frozen snapshot and lineage files are archived at `https://doi.org/10.5281/zenodo.23464809`. The code, pre-registration, manifest and timestamp proof are at `https://github.com/owizdom/pollen`, with the lock at tag `prereg-v1`.

References

[1] Norman E. Breslow and Nicholas E. Day. *Statistical Methods in Cancer Research, Volume 1: The Analysis of Case-Control Studies*. International Agency for Research on Cancer, 1980.

[2] Rebecca DerSimonian and Nan Laird. Meta-analysis in clinical trials. *Controlled Clinical Trials*, 7(3):177–188, 1986.

[3] David Firth. Bias reduction of maximum likelihood estimates. *Biometrika*, 80(1):27–38, 1993.

[4] Andi Gu, Alek H. Kemeny, Susanne F. Yelin, and Mikhail D. Lukin. Discovering state-of-the-art quantum circuits with large language models, 2026. Version 1, 15 October 2026.

[5] Jr. Harrell, Frank E. *Regression Modeling Strategies*. Springer, 2 edition, 2015.

[6] Joachim Hartung and Guido Knapp. A refined method for the meta-analysis of controlled clinical trials with binary outcome. *Statistics in Medicine*, 20(24):3875–3889, 2001.

[7] Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979.

[8] Sreeram Kannan. Post on X by @sreeramkannan, 16 october 2026, 2026. Text as quoted in the introduction.

[9] Roger Koenker and Jr. Bassett, Gilbert. Regression quantiles. *Econometrica*, 46(1):33–50, 1978.

[10] Kydo. Reply on X by @0xkydo, 17 october 2026, 2026. Text as quoted in the introduction.

[11] Jieyi Long. ecdsafail-supplemental-materials. `https://github.com/jieyilong/ecdsafail-supplemental-materials`, 2026. Commit ef7ecd2, 28 August 2026.

[12] Jieyi Long, Theodore Pender, Zhao Huang, Manuel B. Santos, Samrendra Kumar Singh, Bartosz Naskręcki, Bit Wonka, Pierre-Luc Dallaire-Demers, Francesco Giannicola, Ruben M. L. Paschoarelli, Oli Freuler, Jackie Chia-Hsun Lee, Vasily Gnuchev, Gopi Kannappan, John Boyer, Xavier Butler, Akash Balasubramani, Jordan Newman, Bereket Dereje, Alexander Hertlein, Robert Kodra, Lucas Levy, Shaan Patel, JT Rose, Matt Zweil, Wisdom Okechukwu, Tarek El-Eter, Edison Lee, Michael Dong, Alan Li, Anto Joseph, Gajesh Naik, Gautham Anant, Soubhik Deb, and Justin Drake. ECDSA.Fail: Open autoresearch for optimizing elliptic-curve point addition in Shor's algorithm, 2026. Version 2, 19 September 2026.

[13] Donald J. Schuirmann. A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability. *Journal of Pharmacokinetics and Biopharmaceutics*, 15(6):657–680, 1987.

[14] Kurex Sidik and Jeffrey N. Jonkman. A simple confidence interval for meta-analysis. *Statistics in Medicine*, 21(21):3153–3159, 2002.

[15] Peter Todd. OpenTimestamps. `https://opentimestamps.org`, 2016.

[16] Wolfgang Viechtbauer. Confidence intervals for the amount of heterogeneity in meta-analysis. *Statistics in Medicine*, 26(1):37–52, 2007.

[17] Yukon. Yukon: the platform for open frontier research. `https://yukon.org`; public API `https://api.yukon.org`, 2026. Accessed 18 October 2026.

[18] Yukon. Yukon CLI skill: submission, attribution and sync rules, 2026. Agent skill file distributed with the Yukon CLI, accessed 18 October 2026.

A Implementation details

This appendix describes code behaviour that the main text summarises.

A.1 Snapshot fetch and loader

The fetch script, `snapshot/fetch.py`, spaces general requests at no more than eight per second and attribution calls at no more than two starts per second, with six attribution calls in flight. It retries rate-limited calls within a fifteen-minute budget, and gives server errors and timeouts up to five attempts. The loader, `snapshot/load.py`, never touches the network. Each time it opens a file it checks the file against `SHA256SUMS`, checks `FREEZE.json` against its own sum, and checks the row count against `FREEZE.json`. It also refuses a listed file that is missing and a data file that is not listed.

A.2 Tree recovery and candidate parents

The build in `lineage/build.py` fetches every tree commit by its hash into a local bare mirror, never by branch name. A failing batch of hashes is split in halves until the unfetchable hashes are isolated, and those are written to `missing.json`. The candidates for a child are earlier analysis rows with a tree, collapsed by tree identity so that only the most recent row with each tree remains. From them the build keeps the 50 most recent promoted rows whose promotion finished before the child was created. A solver's own unpromoted rows are not candidates; an earlier rule that admitted them failed AT4 before the lock (Section 3.4).

A.3 Lock check

The lock is a manifest, `docs/prereg-manifest.json`. It holds the SHA-256 of the pre-registration text, every code file under `snapshot/`, `families/`, `lineage/`, `outcomes/` and `analysis/`, the family table, the denominator classes, and the checksum files of the frozen snapshot and lineage data. The lock check in `analysis/lock.py` recomputes every hash and verifies every frozen data file against its checksum file. It rejects unlisted data files, symbolic links and compiled files without source. It also requires the manifest to equal the copy under the git tag `prereg-v1`. Every analysis driver runs the check first, and the confirmatory script exits with an error if it fails. The equivalence margin is read from a single line of `docs/prereg.md`.

A.4 Checks on the estimator

Two automated tests check the pipeline on simulated data with the real treatment structure. In AT13, outcomes with a planted odds ratio of 1.5 must be recovered inside the 95% interval in at least 90 of 100 fixed seeds. A correct 95% interval falls below 90 of 100 with probability about 0.03 under a Binomial(100, 0.95) count. In AT14, shuffling family within solver must give an H1 interval that covers 1 in at least 90 of 100 fixed seeds. At the lock, coverage was 92 of 100 for AT13 and 97 of 100 for AT14.

B Secondary tests

Table 4 lists every secondary test from `results/secondary.json`. Events are outcome events in the smaller arm; a test enters the Holm family (C) only with at least 20, and is otherwise descriptive (D). Estimates are log odds ratios for H1, H2 and H5, the share of cross-family promotions that are large jumps for H3a, and the median-regression coefficient on log gain for H4. H3b, plateau breaks, had at most one event in the smaller arm on every benchmark, and all its tests are descriptive; its rows are omitted.

Table 4: Secondary tests. A dash marks a value that does not exist, such as an adjusted p for a descriptive test or an estimate for an unidentified fit.

Test	Benchmark	Events	Estimate	p	Holm p	Status
H1	ecdsafail-challenge	56	-0.897	0.105	1.000	C
H2	mlxfast-gemma4-26b-a4b	0	1.017	0.550	–	D
H2	qwen38-125b-a6b-cuda-v1	0	–	–	–	D
H2	eip8200-challenges/modexp	0	-0.523	0.712	–	D
H2	eip8200-challenges/ripemd160	5	-0.014	0.992	–	D
H2	flock-challenge	0	-1.289	0.259	–	D
H2	flock-challenge-multi/x86	5	1.776	0.023	–	D
H2	lighter-prover-challenge	2	-0.397	0.591	–	D
H2	mlxfast-challenge	2	0.479	0.814	–	D
H2	sig-golf/full	3	0.458	0.792	–	D
H2	ecdsafail-challenge	8	-1.970	0.236	–	D
H3a	mlxfast-gemma4-26b-a4b	7	0.072	0.553	–	D
H3a	qwen38-125b-a6b-cuda-v1	3	0.091	1.000	–	D
H3a	eip8200-challenges/modexp	2	0.041	1.000	–	D
H3a	eip8200-challenges/ripemd160	6	0.053	0.105	–	D
H3a	flock-challenge	10	0.112	0.813	–	D
H3a	flock-challenge-multi/x86	2	0.079	1.000	–	D
H3a	lighter-prover-challenge	4	0.051	0.203	–	D
H3a	mlxfast-challenge	2	0.089	1.000	–	D
H3a	sig-golf/full	6	0.101	0.621	–	D
H3a	ecdsafail-challenge	1	0.078	0.145	–	D
H4	mlxfast-gemma4-26b-a4b	57	-0.000	0.738	1.000	C
H4	qwen38-125b-a6b-cuda-v1	37	0.001	0.140	1.000	C
H4	eip8200-challenges/modexp	56	0.000	0.790	1.000	C
H4	eip8200-challenges/ripemd160	79	-0.000	0.013	0.154	C
H4	flock-challenge	79	-0.000	0.705	1.000	C
H4	flock-challenge-multi/x86	28	-0.000	0.797	1.000	C
H4	lighter-prover-challenge	53	0.002	0.246	1.000	C
H4	mlxfast-challenge	34	0.001	0.560	1.000	C
H4	sig-golf/full	73	0.000	0.847	1.000	C
H4	ecdsafail-challenge	56	-0.000	0.653	1.000	C
H5	ecdsafail-challenge	102	-0.821	0.234	1.000	C